



YiXiang Shen

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Loc: Nanjing

Age: 20



<<< Education & Research Interests

2023-now

JNU/Dept. of Mathematics

Major: Information and Computing Science

GPA: 4.44/5 (rank 1/39)

Eng. level: CET-6 (570)

Core course: Analysis I/II/III (90/94/100), Algebra I/II (97/96), Probability and Statistics (98), ODE (99), Database (97), Intelligent Computing (97), Operational research (100), Data Structure (100), Real Analysis (98), Machine Learning (98)

Research Interests: The application of computational mathematics in machine learning and AI algorithms; deep learning; causal inference; statistics; reinforcement learning

<<< Learning Experience

2025-2026

College Students' Innovative Entrepreneurial Training Plan Program

Member

[Backgrounds] Addressing the issues of poor transmission performance, low transmission rate, and security concerns in the image transmission process of drone low-altitude logistics, a collaborative optimization strategy is constructed, incorporating multimodal data semantic information extraction (UT+CNN), edge computing (Ascend AI), and secure encoding (DeepJSCEC).

[Role] This project has been approved at the national level and completed with an excellent rating. As a core member of the team, I was responsible for developing the front-end interface for drone control and designing the user command input module, which has obtained a software copyright registration certificate. I was responsible for drawing charts and diagrams, as well as proofreading and formatting the entire documents.

2026.05

Research on Causal Inference

Leader

[Role] Based on causal representation learning and structural causal models, and drawing on the backdoor adjustment idea in the CROCODILE framework, a lightweight and stably trainable causal query classification model was designed. The model uses ResNet50 to extract shared features, aggregates category-level disease evidence through lightweight feature projection, a Transformer encoder, and a task query mechanism, and introduces intra-batch permutation and Dropout to construct intervention representations to achieve approximate training through backdoor adjustment. Building on this, this paper further explores the complete framework and analyzes the main sources of training instability through ablation experiments.

2025/2024

Mathematical Modeling Competition

Leader

[Role] (1) Responsible for data collection, cleaning, and preprocessing; conducted in-depth analysis and interpretation, proposed directions for model improvement, such as introducing Bayes hierarchical structure and time-varying elasticity function to enhance prediction accuracy and robustness; independently wrote English papers, ultimately completing 1 paper.

(2) Under the poor results of multiple linear regression, I led the construction of a systematic risk prediction system based on logistic regression and NAR neural network. During the modeling process, I introduced deep learning algorithms to optimize traditional models, enhancing their robustness and prediction accuracy in complex market environments.

[Prize] Second prize, Undergraduate group; (2) Third prize, Undergraduate group

<<< Honors

(1) National Scholarship (2024-2025)

(3) Second-class scholarship for outstanding students of JNU (2023-2024)

(2) Outstanding Young Volunteer of JNU (2024-2025)

(4) Outstanding staff member and department head of the college student union (2023-2024)

<<< Self-assessment & Skills

· C/C++, MATLAB, Python, and related libraries, Tensorflow, LLMs

· Familiar with the general process of project work, with team practical experience ranging from problem comprehension, data cleaning, model construction and optimization to result analysis

> As a student pursuing interdisciplinary development, I embrace a lifelong learning mindset and am willing to embrace new content and knowledge.